

MEASURING AND ENHANCING TRUSTWORTHINESS OF LLMs IN RAG

Through Grounded Attributions and Learning to Refuse

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- TRUST-SCORE: holistic metric for LLM-in-RAG trustworthiness across answers, citations, and refusals.
- Prompting (e.g., in-context learning) under-adapts models; we introduce TRUST-ALIGN to align behavior.
- 26/27 model–dataset configs beat baselines on ASQA, QAMPARI, and ELI5.

+12.56

LLaMA-3-8b vs FRONT on ASQA (TRUST-SCORE)

+36.04

vs FRONT on QAMPARI (TRUST-SCORE)

+17.69

vs FRONT on ELI5 (TRUST-SCORE)

PAC: Precision over Answered Cases

RAC: Recall over Answerable Cases

AR%: Answered Ratio

Method (TRUST-ALIGN) + Metrics (TRUST-SCORE)

- Task: given question q and documents D , generate statements with inline citations; refuse if D is insufficient.
- TRUST-ALIGN: construct 19K preference pairs covering five error types; align via Direct Preference Optimization (DPO).
- Metrics (first-use expansions on this slide):
 - Answer Correctness F1 (F1AC): combines PAC (Precision over Answered Cases) and RAC (Recall over Answerable Cases).
 - Grounded Refusals F1 (F1GR): F1 over correctly refusing unanswerables and answering answerables.
 - Grounded Citations F1 (F1GC): harmonic mean of citation precision (Pcite) and citation recall (Rcite).
 - Answered Ratio (AR%): fraction of questions answered.
- $\text{TRUST-SCORE} = (\text{F1GR} + \text{F1AC} + \text{F1GC}) / 3$.

Figure 1 — TRUST-SCORE computational graph (PAC, RAC, refusal/answer, citation NLI)

Figure 2 — TRUST-ALIGN data curation and DPO pipeline

PAC: Precision over Answered Cases

RAC: Recall over Answerable Cases

Pcite/Rcite: citation precision/recall

AR%: Answered Ratio

Results

- TRUST-ALIGN improves TRUST-SCORE in 26/27 settings; DPO generally outperforms SFT on TRUST-SCORE and Grounded Citations F1 (F1GC).
- LLaMA-3-8b vs FRONT (TRUST-SCORE): +12.56 (ASQA), +36.04 (QAMPARI), +17.69 (ELI5).
- Out-of-domain (ExpertQA): TRUST-ALIGNed LLaMA-3-8b achieves 54.85 vs GPT-4 in-context learning 54.69.

26/27
Configurations where TRUST-ALIGN wins

↑ F1GC
Grounded Citations F1 (citation precision/recall)

↑ F1GR
Grounded Refusals F1 (refuse unanswerables, answer answerables)

F1AC: Answer Correctness F1

F1GR: Grounded Refusals F1

F1GC: Grounded Citations F1

AR%: Answered Ratio

Findings

- Many state-of-the-art LLMs overuse parametric knowledge in RAG, lowering TRUST-SCORE; prompting can cause over-refusal or over-responsiveness.
- TRUST-ALIGN raises Grounded Refusals F1 (F1GR) and Grounded Citations F1 (F1GC) while moderating Answered Ratio (AR%).
- Generalizes across LLaMA, Qwen, and Phi3.5 families; competitive with GPT-4 on ExpertQA.

F1AC = Answer Correctness F1

F1GR = Grounded Refusals F1

F1GC = Grounded Citations F1

AR% = Answered Ratio

Closing

- TRUST-SCORE averages Answer Correctness F1 (F1AC), Grounded Refusals F1 (F1GR), and Grounded Citations F1 (F1GC) to isolate LLM behavior from retrieval.
- TRUST-ALIGN builds 19K preference pairs across hallucination types and aligns models via Direct Preference Optimization (DPO).
- Consistent gains across ASQA, QAMPARI, and ELI5; strong out-of-domain performance on ExpertQA.

F1AC: Answer Correctness F1

F1GR: Grounded Refusals F1

F1GC: Grounded Citations F1

AR%: Answered Ratio